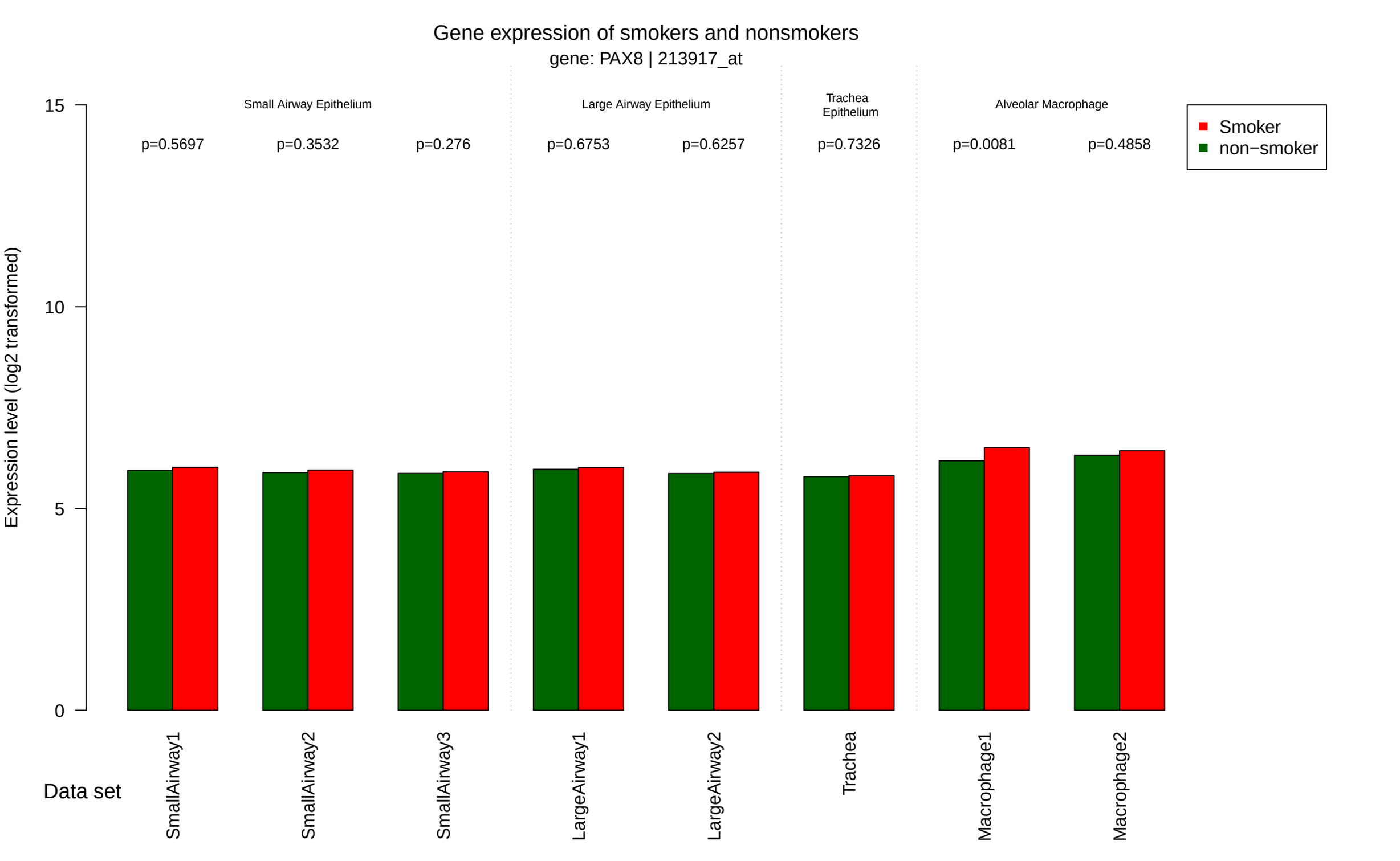
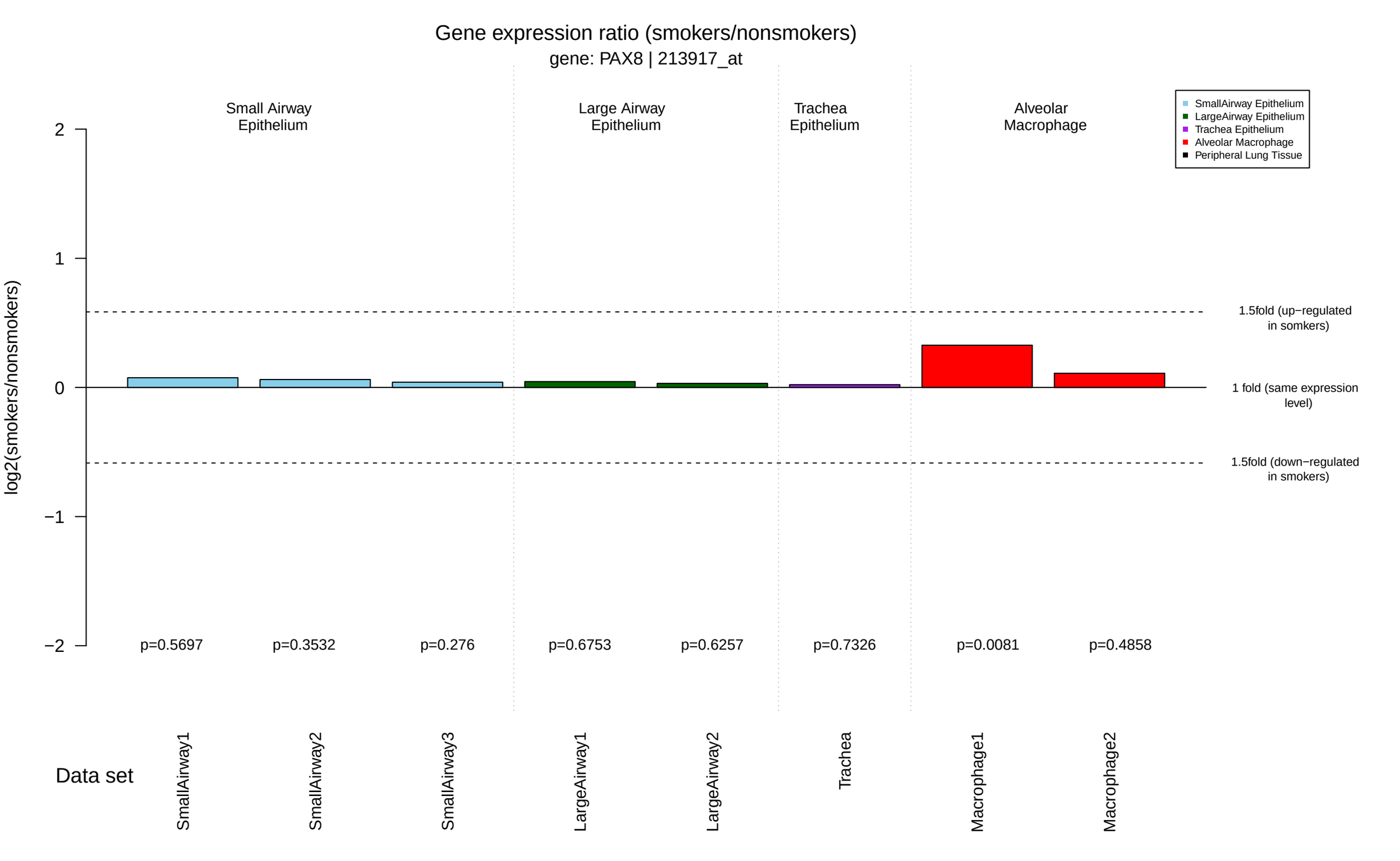
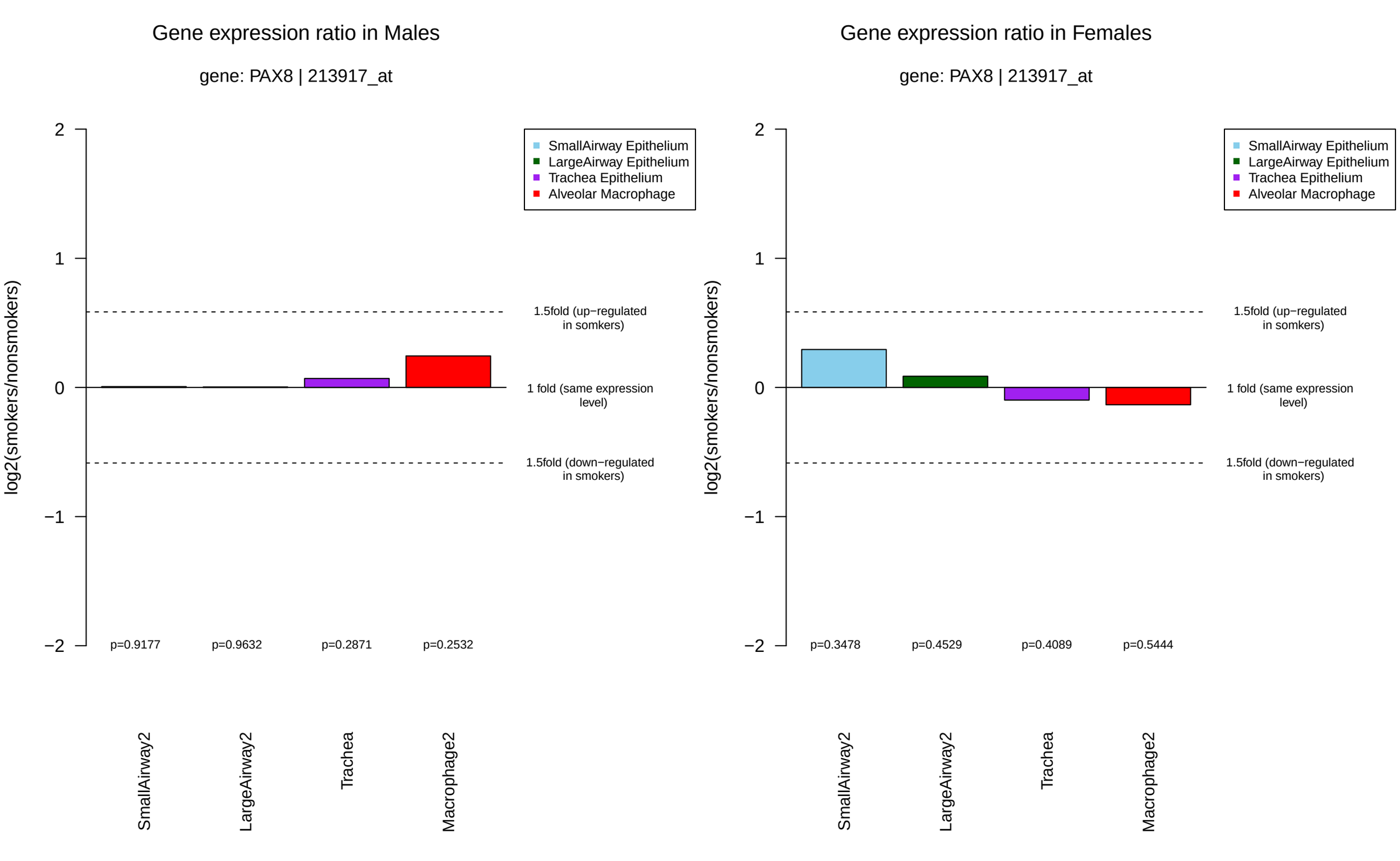
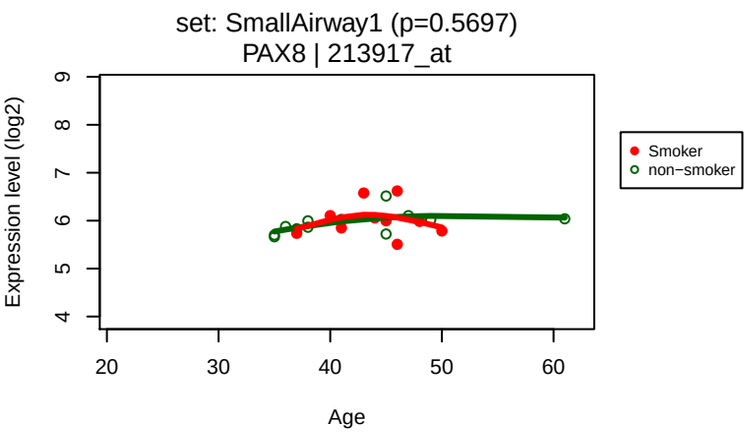
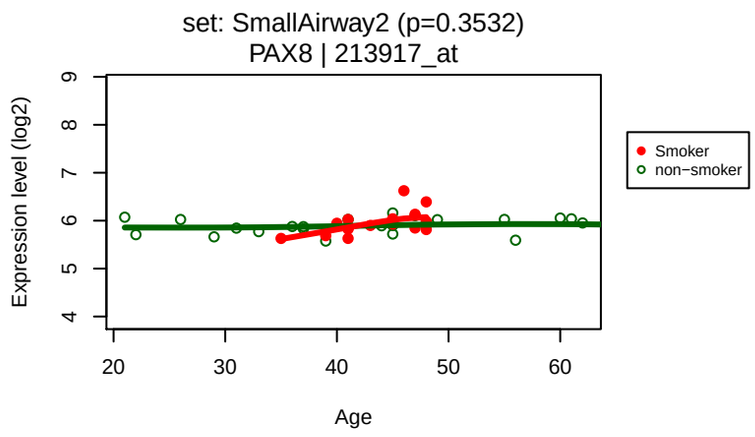
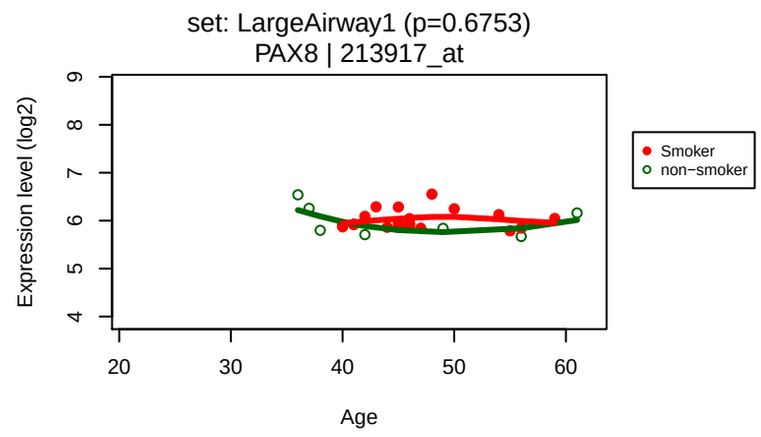

Gene expression of smokers and nonsmokers
gene: PAX8 | 213917_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5697
p=0.3532
p=0.276
p=0.6753
p=0.6257
p=0.7326
p=0.0081
p=0.4858
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

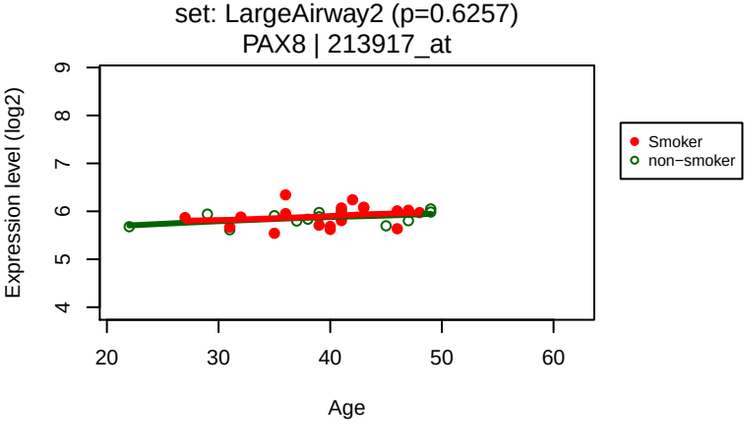

Gene expression ratio (smokers/nonsmokers)
gene: PAX8 | 213917_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5697
p=0.3532
p=0.276
p=0.6753
p=0.6257
p=0.7326
p=0.0081
p=0.4858
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

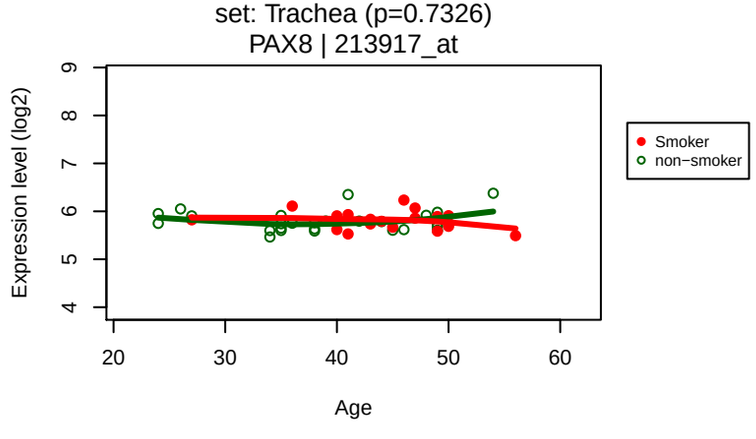

Gene expression ratio in Males
gene: PAX8 | 213917_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9177
p=0.9632
p=0.2871
p=0.2532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX8 | 213917_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3478
p=0.4529
p=0.4089
p=0.5444
SmallAirway2
LargeAirway2
Trachea
Macrophage2

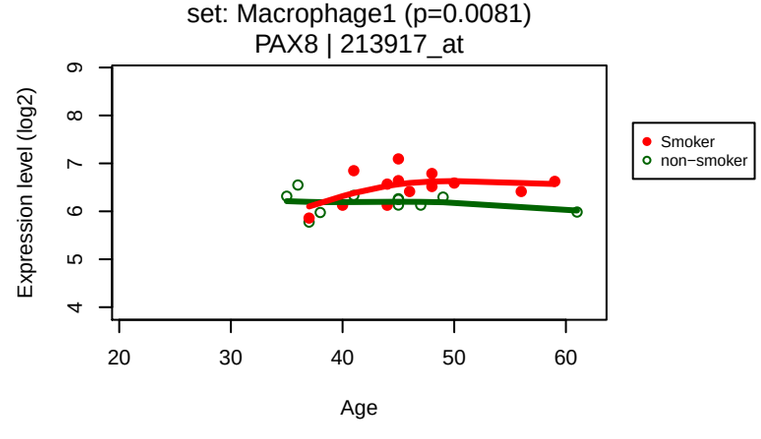

set: SmallAirway1 (p=0.5697)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker

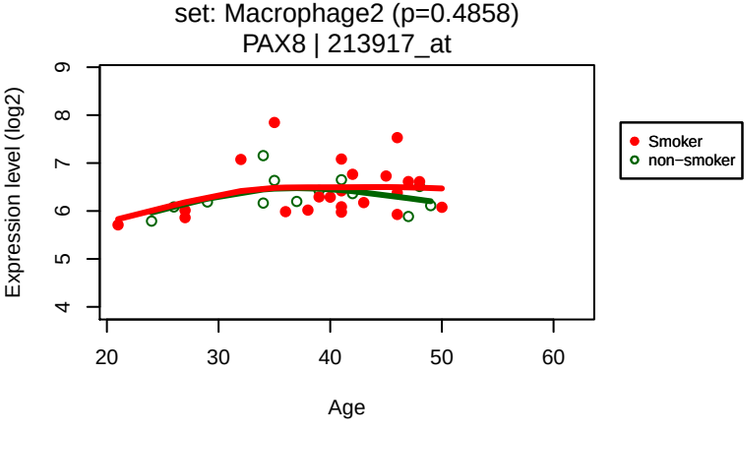

set: SmallAirway2 (p=0.3532)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker

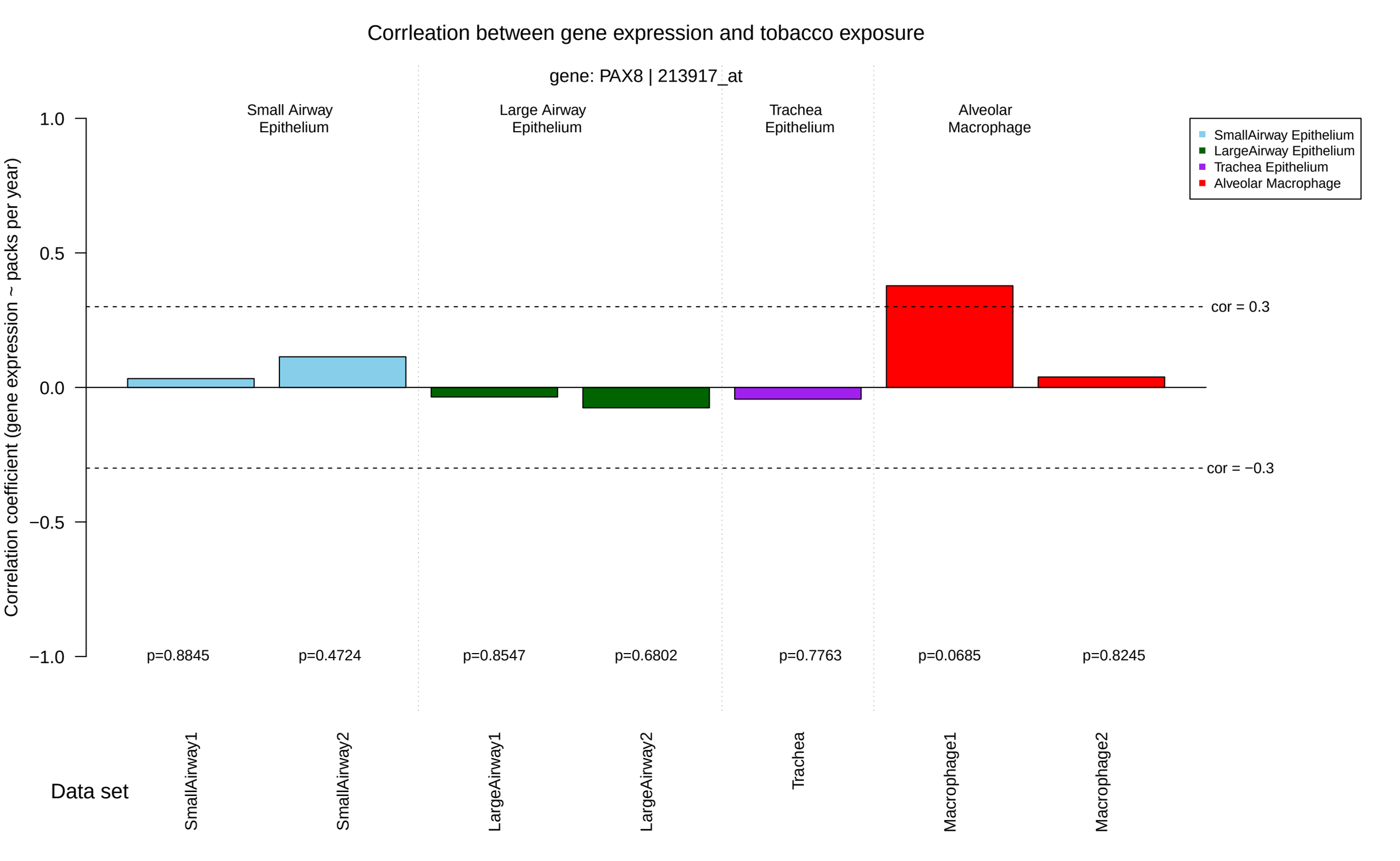

set: LargeAirway1 (p=0.6753)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6257)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7326)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0081)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4858)
PAX8 | 213917_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX8 | 213917_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8845
p=0.4724
p=0.8547
p=0.6802
p=0.7763
p=0.0685
p=0.8245
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2